PubMed x Gemini Notebook x Claude Code MCP



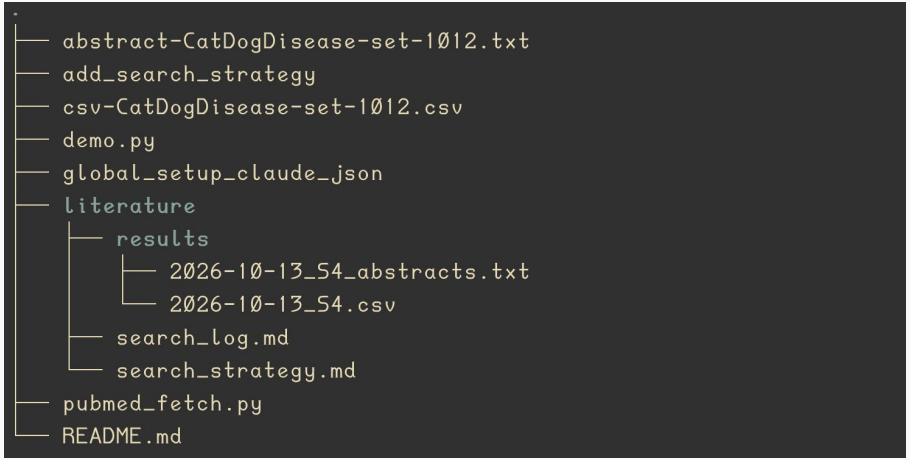
Agenda

- PubMed Queries
- Gemini Notebook
- Claude Code MCP & Automatic Script

Before We Start...

https://github.com/yienlyu/research_project_demo.git

```
> git clone https://github.com/yienlyu/research_project_demo.git
```



Or download ZIP

1

Finding papers…

PubMed Queries

https://pubmed.ncbi.nlm.nih.gov/



PubMed Queries

- What we want to find: (貓 OR 狗) AND (淋巴癌) AND (化療)

- Possible keywords
  - 貓：Cat、Felidae
  - 狗：Dog、Canidae
  - 淋巴癌：Lymphoma、Lymphosarcoma、Lymph Tumor、Lymph Cancer
  - 化療：Chemotherapy、Chemotherapeutic protocols、Chemotherapeutic management

PubMed Queries

- The query:

  ((Cat OR Felidae) OR (Dog OR Canidae)) AND (Lymphoma OR Lymphosarcoma OR “Lymph Tumor” OR “Lymph Cancer”) AND (Chemotherap*)

- Limit the field: (keyword [field tag])
  - Lymphoma [TI] → Title
  - Dog [tiab] → Title/Abstract
  - UCLA [ad] → Affiliation

- Search field tags: https://pubmed.ncbi.nlm.nih.gov/help/#using-search-field-tags

Quick Tips: Truncation

- chemotherap* → chemotherapy, chemotherapies, chemotherapeutic
- At least 4 characters before the first *: colo* works, co* doesn't
- Multiple wildcards in one term are allowed: organi*ation*

Quick Tips: Truncation

- But * turns OFF **Automatic Term Mapping** – and no MeSH explosion

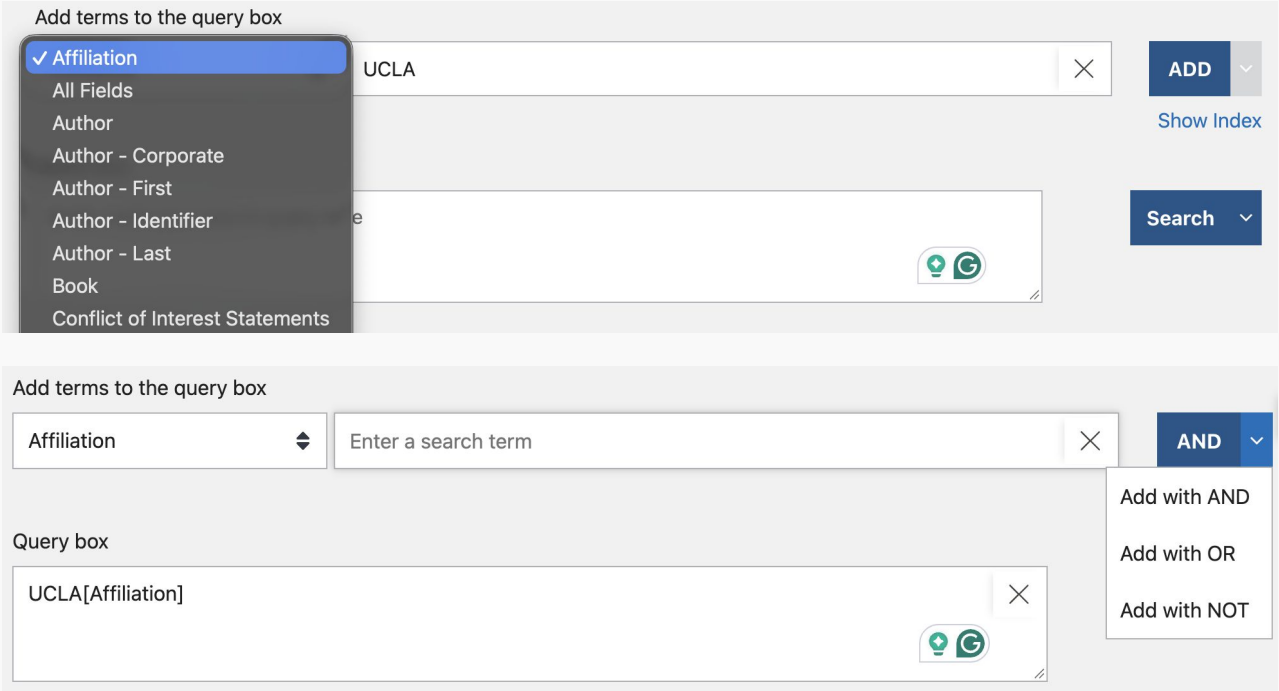
You'd want to write	What it actually catches
cat*	catheter, catabolism, category, catalyst, …
dog*	dogma, dogmatic, …
feline*	safe
canine*	safe

PubMed Queries



PubMed Queries

https://pubmed.ncbi.nlm.nih.gov/advanced/



PubMed Queries

History and Search Details Download Delete

Search	Actions	Details	Query	Results	Time
#4	•••	›	Search: **(Lymphoma[title] OR Lymphosarcoma[title]) AND (((Cat OR Felidae) OR (Dog OR Canidae)) AND (Lymphoma OR Lymphosarcoma OR "Lymph Tumor" OR "Lymph Cancer") AND (Chemotherap*))**	718	01:26:54
#3	•••	›	Search: **(Lymphoma[title] OR Lymphosarcoma[title]) OR (((Cat OR Felidae) OR (Dog OR Canidae)) AND (Lymphoma OR Lymphosarcoma OR "Lymph Tumor" OR "Lymph Cancer") AND (Chemotherap*))**	120,581	01:26:34
#2	•••	›	Search: **Lymphoma[title] OR Lymphosarcoma[title]**	120,310	01:22:54
#1	•••	›	Search: **((Cat OR Felidae) OR (Dog OR Canidae)) AND (Lymphoma OR Lymphosarcoma OR "Lymph Tumor" OR "Lymph Cancer") AND (Chemotherap*)**	989	00:24:37

PubMed Queries
MeSH (Medical Subject Headings)

- MeSH Database: https://www.ncbi.nlm.nih.gov/mesh

Search	Actions	Details	Query	Results	Time
#5	•••	⌄	Search: **cancer** "cancer s"[All Fields] OR "cancerated"[All Fields] OR "canceration"[All Fields] OR "cancerization"[All Fields] OR "cancerized"[All Fields] OR "cancerous"[All Fields] OR "neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "cancer"[All Fields] OR "cancers"[All Fields]	5,705,052	01:39:53
			Translations **cancer:** "cancer's"[All Fields] OR "cancerated"[All Fields] OR "canceration"[All Fields] OR "cancerization"[All Fields] OR "cancerized"[All Fields] OR "cancerous"[All Fields] OR "neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "cancer"[All Fields] OR "cancers"[All Fields]		

PubMed Queries
Automatic Term Mapping (ATM)

- What you type is not what PubMed searches!

You wrote	What happens
“heart attack” in quotes	No mapping, exact phrase only
chemotherap* wildcard	No mapping
Dog[tiab] field tag	No mapping, that field only

PubMed Queries
MeSH

Which is the best query strategy?

a. “aspirin for heart attack prevention”

b. aspirin heart attack prevention

c. aspirin AND heart AND attack AND prevention

PubMed Queries
MeSH

- Which is the best query strategy?

a. ~~“aspirin for heart attack prevention”~~ Quotation marks turn off Automatic Term Mapping – you just disabled its best feature

b. aspirin heart attack prevention

Add nothing, and PubMed maps it against MeSH for you

c. ~~aspirin AND heart AND attack AND prevention~~ Splitting `heart` and `attack` loses "myocardial infarction" as a concept

PubMed Queries
MeSH

A Anatomy
B Organisms
C Diseases
D Chemicals and Drugs
E Analytical, Diagnostic and Therapeutic Techniques…
F Psychiatry and Psychology
G Phenomena and Processes
H Disciplines and Occupations
I Anthropology, Education, Sociology…
J Technology, Industry, Agriculture
K Humanities
L Information Science
M Named Groups
N Health Care
V Publication Characteristics
Z Geographicals

MeSH Term - Tree

Cats - **B** Organisms
Cat Diseases - **C22.180**

C Diseases
-> **22** Animal Diseases
-> **180** Cat Diseases

PubMed Queries
MeSH

- **Dog Diseases**: Diseases of the domestic dog (Canis familiaris). This term does not include diseases of wild dogs, WOLVES; FOXES; and other Canidae.

- Explosion? – [MeSH:NoExp]

- All MeSH categories: https://www.ncbi.nlm.nih.gov/mesh/1000048

2

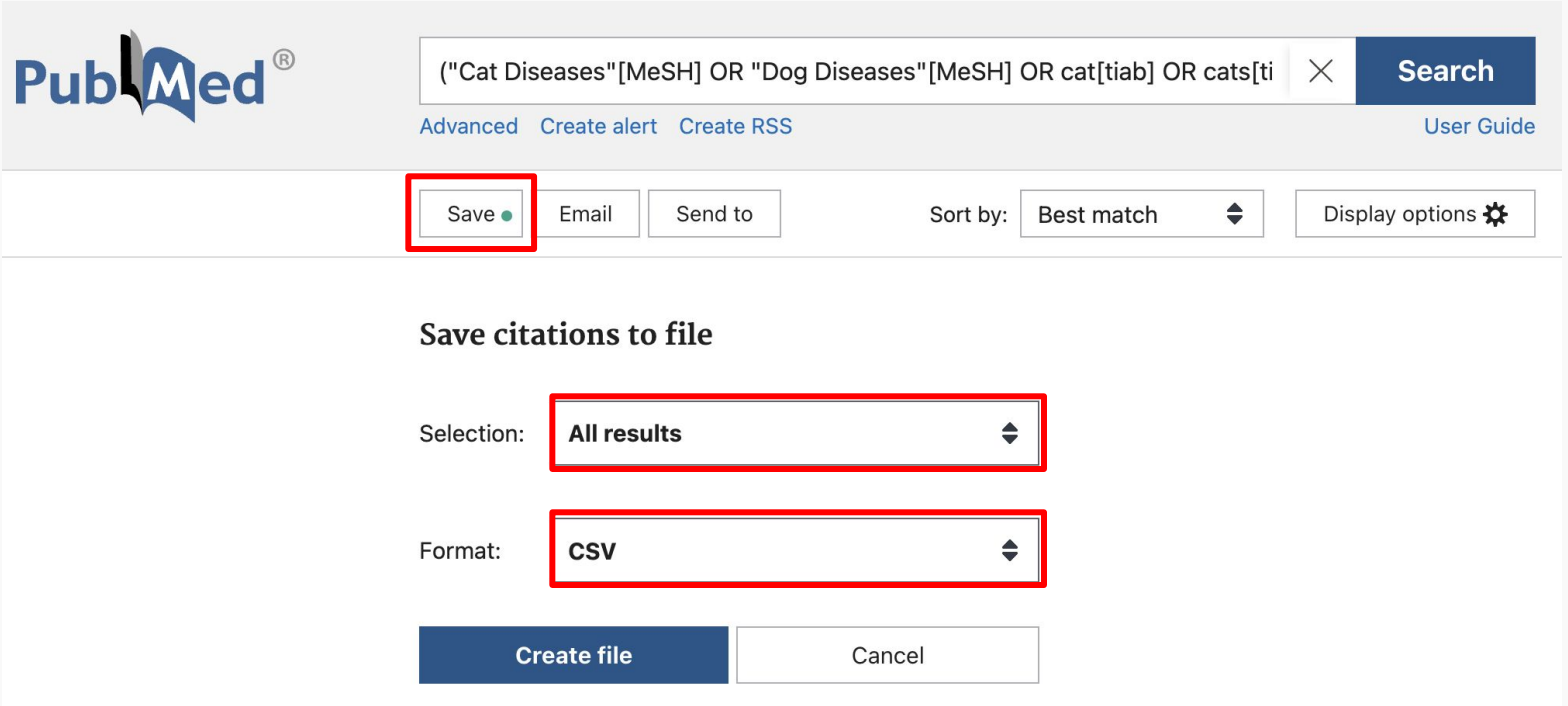
Gemini Notebook

Read the Papers Quick and Nicely

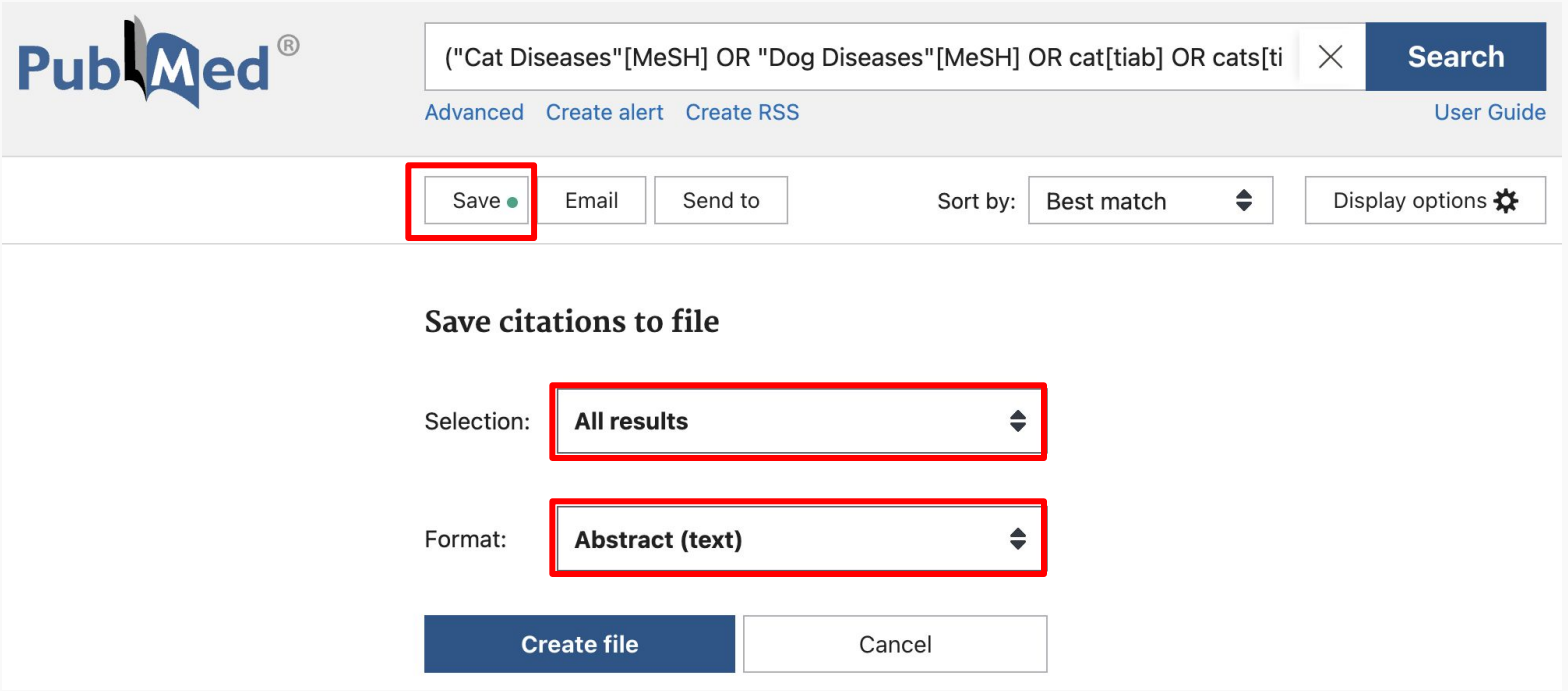
The queries for today's example:

```
("Cat Diseases"[MeSH] OR "Dog Diseases"[MeSH] OR cat[tiab] OR
cats[tiab] OR feline*[tiab] OR dog[tiab] OR dogs[tiab] OR
canine*[tiab]) AND ("Lymphoma"[MeSH] OR lymphoma*[tiab] OR
lymphosarcoma*[tiab]) AND ("Antineoplastic Combined
Chemotherapy Protocols"[MeSH] OR "Antineoplastic
Agents"[MeSH] OR chemotherap*[tiab] OR "CHOP protocol"[tiab])
```

Download from Site: to csv



Download from Site: to txt



Read the Papers Quick and Nicely

Make good use of Gemini Notebook!

- Answer ONLY from your uploaded sources
- Every answer cites, and the citation jumps back to the text
- Won't invent what you didn't give it

```
Save -> Abstract (text) -> .txt -> to Gemini Notebook <- for reading!
Save -> CSV             -> .csv -> to track           <- for management!
```

How to Prompt

Pattern	Ask it like this (cat/dog lymphoma)	Why it works
Compare	What **different** outcome measures did these studies use for treatment response?	Forces the differences between sources into view
Evidence	Which paper best supports a causal claim, and **on what grounds**?	Makes it show its reasoning so you can judge
Gap	What question do these sources **all fail to answer**?	Its strongest question, and your weakest

Sources

Add sources

Search the web for new sources

Web Fast Research

Select all

abstract-CatDogDisease-set-1...



research_project_demo

1 source · Oct 12, 2026

The provided literature primarily explores the **diagnosis, classification, and treatment of lymphoma** in companion animals such as dogs and cats. Multiple studies evaluate the efficacy and safety of various **chemotherapy protocols**, including CHOP-based regimens, rescue treatments for relapsed cases, and novel drug combinations aimed at improving remission and survival rates. Additionally, researchers examine **prognostic factors** like immunophenotype, microRNA expression, and biochemical markers to better predict patient outcomes. The sources also highlight potential **adverse effects and toxicity profiles** associated with antineoplastic therapies, alongside the mechanisms of **drug resistance** that complicate cancer management in veterinary oncology.

Save to note

Ask a question or create something 1 source

Gemini Notebook can make mistakes, so double-check it.

Studio

Audio Overvi... Slide Deck

Video Overvi... Mind Map

Reports New! Flashcards

Quiz Infographic

Data Table

Studio output will be saved here.

After adding sources, click to add Audio Overview, Study Guide, Mind Map, and more!

Add note

Prompt

Example:

“”””

For each source, list:

research question / method and sample / main findings / limitations

Tag every item with its source number.

If a source doesn't state something, write "not reported." Don't infer.

“”””

Sources

Add sources

Search the web for new sources

Web Fast Research

Select all

abstract-CatDogDisease-set-1...

Source 1: Bayer AL et al. (2024) 1

- **Research Question**: How does doxorubicin treatment induce an aberrant immune response (specifically CD8+ T cells) and drive cardiotoxicity, adverse cardiac remodeling, and systolic dysfunction? 1 2
- **Method and San** doxorubicin, cell cohort evaluation ligands in canine hematopoietic m
- **Main Findings**: D IFNγ+CD8+ T cell depletion of CD8 systolic function cardiac fibroblast granzyme B relea human cancer pa cells post-treatm CXCL9 and CXCL
- **Limitations**: not reported. 2

abstract-CatDogDisease-set-1012.txt

Doxorubicin, the most prescribed chemotherapeutic drug, causes dose-dependent cardiotoxicity and heart failure. However, our understanding of the immune response elicited by doxorubicin is limited. Here we show that an aberrant CD8+ T cell immune response following doxorubicin-induced cardiac injury drives adverse remodeling and cardiomyopathy. Doxorubicin treatment in non-tumor-bearing

View source

Ask a question or create something 1 source

Gemini Notebook can make mistakes, so double-check it.

Studio

Audio Overvi... Slide Deck

Video Overvi... Mind Map

Reports New! Flashcards

Quiz Infographic

Data Table

Studio output will be saved here.

After adding sources, click to add Audio Overview, Study Guide, Mind Map, and more!

Add note

Prompt (cont.)

Example:

“””

Make a table of all sources with these columns:

source number | species of the study subjects | human or animal | study type

One row per source. If there are no living subjects, write "in vitro"

in the species column. If not stated, write "not reported."

“””

Prompt (cont.): why “Cat Diseases”[MeSH]?

	amount
dog	17
cat	2 (source 5, 14)
human	8
rat	5
rabbit	2 (source 2, 16)
In vitro (no living subjects)	7
No dog nor cat	14

Build the Matrix: For Gemini Notebook

“””

Organize all sources into a table:

first author | year | study design | sample size |

main finding | limitations the authors state

One row per paper, tagged with its source number.

If not mentioned, write "not reported."

“””

Build the Matrix: For You

Why I kept this paper?

3

PubMed Reproducible Search

Before We Start...

https://github.com/yienlyu/research_project_demo.git

```
> git clone https://github.com/yienlyu/research_project_demo.git
```

```
.
├── abstract-CatDogDisease-set-1012.txt
├── add_search_strategy
├── csv-CatDogDisease-set-1012.csv
├── demo.py
├── global_setup_claude_json
├── literature
│   ├── results
│   │   ├── 2026-10-13_S4_abstracts.txt
│   │   └── 2026-10-13_S4.csv
│   ├── search_log.md
│   └── search_strategy.md
├── pubmed_fetch.py
└── README.md
```

Can We Reproduce the Exact Same Result?

	How long it lasts	Limits
Search History	**Expires after 8 hours idle**	Last 100 searches
Clipboard	**Expires after 8 hours idle**	500 items
Collections (sign-in required)	Permanent	1,000 per collection
Files you downloaded	Yours	-

Can We Reproduce the Exact Same Result?

❶ **Scratchpad**: PubMed history

❷ **Record**: Write it to a file (strategy + log)

❸ **Automation**: Run and write it automatically!

Model Context Protocol (MCP)

- What is MCP? A socket standard
- What can MCP enable? Lets the AI actually query instead of guessing from memory
- Why does MCP matter? Whether it can touch real data

Intro to Terminal





Action	Command
Where am I?	pwd
What's in here?	ls
Create a folder	mkdir folder_name
Go into a folder	cd folder_name
Go up one level	cd ..

Claude Code MCP

- env setup

```
# install LTS Node.js
brew install node

# check the version of Node.js and npx
Node -v  # v20.x.x or v22.x.x
npx -v
```

Claude Code MCP

- test

```
npx -y @ncukondo/mcp-server-pubmed --email your_email@example.com
```

- Change to your email address

  -> no ERROR and stay on the pending status -> OK!

- End the test: press `Ctrl + C`

Claude Code MCP

- Write a global setup file `~/.claude.json`
- Remember to change your email!
- Find the text file in repo `global_setup_claude_json`

```
python3 -c '
import json, os

path = os.path.expanduser("~/.claude.json")
try:
    with open(path, "r") as f:
        data = json.load(f)
except Exception:
    data = {}

if "mcpServers" not in data:
    data["mcpServers"] = {}

data["mcpServers"]["pubmed"] = {
    "command": "npx",
    "args": [
        "-y",
        "@ncukondo/mcp-server-pubmed",
        "--email",
        "your_email@example.com"
    ]
}

with open(path, "w") as f:
    json.dump(data, f, indent=2)

'
```

Claude Code MCP

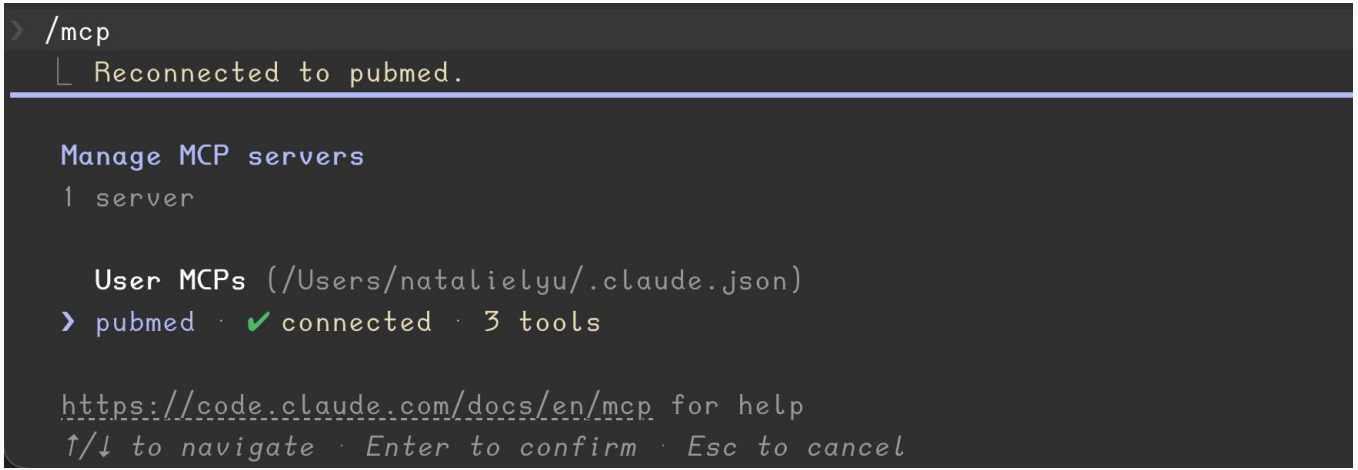
- Verify the connection

```
> cd ~/your_workspace/your_project/
> claude
```

Claude Code MCP

- Entered the interface to check the connection

```
/mcp
```



NotebookLM MCP

```
> uv tool install --python 3.12 notebooklm-mcp-cli
> nlm login --storage protected
> nlm doctor

> nlm setup add claude-code
> claude mcp list
```



NotebookLM MCP

```
> cd ~/your_workspace/your_project/
> nlm notebook create "Cat & Dog Lymphoma"
> nlm alias set lymphoma <notebook-id>
> nlm source add lymphoma --file abstract-CatDogDisease-set-1012.txt
  --wait
> nlm notebook query lymphoma "What species were the study subjects in
  each source?"
```

Claude Code MCP

- Or enter the interface

```
> claude

  - ! 直接跑指令：! nlm notebook query lymphoma "..."
  - 自然語言請 Claude 跑：「執行 nlm notebook query ...」
  - 走 MCP：「用 Gemini Notebook 問 notebook <name / ID>：...」
```

Log setup

- Build a directory for your project

```
# enter your research-only directory
cd ~/your_Workspace/research_project

# build a directory for strategy, log and result storage
mkdir -p literature/results/

# build a search log file
touch literature/search_log.md
```

Search Strategy Setting

- Find the textfile in repo `add_search_strategy`

```
cat << 'EOF' > literature/search_strategy.md
# Literature Search Strategy

- Version: v1.0
- Target Database: PubMed

## Search Blocks

### S1 (Population - Cats and Dogs)
("Cat Diseases"[MeSH] OR "Dog Diseases"[MeSH] OR cat[tiab] OR cats[tiab] OR feline*[tiab] OR dog[tiab] OR dogs[tiab]
OR canine*[tiab])

### S2 (Condition - Lymphoma)
("Lymphoma"[MeSH] OR lymphoma*[tiab] OR lymphosarcoma*[tiab])

### S3 (Intervention - Chemotherapy)
("Antineoplastic Combined Chemotherapy Protocols"[MeSH] OR "Antineoplastic Agents"[MeSH] OR chemotherap*[tiab] OR
"CHOP protocol"[tiab])

### S4 (Combined Query)
("Cat Diseases"[MeSH] OR "Dog Diseases"[MeSH] OR cat[tiab] OR cats[tiab] OR feline*[tiab] OR dog[tiab] OR dogs[tiab]
OR canine*[tiab]) AND ("Lymphoma"[MeSH] OR lymphoma*[tiab] OR lymphosarcoma*[tiab]) AND ("Antineoplastic Combined
Chemotherapy Protocols"[MeSH] OR "Antineoplastic Agents"[MeSH] OR chemotherap*[tiab] OR "CHOP protocol"[tiab])

EOF
```

Enter Claude in Your Local Terminal

Prompt example:

讀取 literature/search_strategy.md，並使用 PubMed MCP 依序執行 S1 到 S4 的檢索式。
嚴格遵守以下規則：
1.【禁止改動】請原封不動送出策略檔中的檢索式，不要擅自優化、增加同義詞或增刪括號。
2.【紀錄 Log】每一條檢索式執行完畢後，將「執行日期」、「策略版本號」、「檢索式編號」、「實際送出的檢索式」與「PubMed 回傳筆數」追加記錄至 literature/search_log.md。
3.【匯出結果檔】請將最終組合檢索式（S4）的所有搜尋結果（包含 PMID、Title、Journal、PubYear、Authors、Abstract）儲存為 CSV 檔案，路徑為：literature/results/20261013_S4.csv。

Enter Claude in Your Local Terminal

Save your prompt at `~/.claude/skills/search-log/SKILL.md`

- - -
name: search-log
description: 讀取 literature/search_strategy.md,並使用 PubMed MCP 依序執行 S1 到 S4 的檢索式。使用者說「search pubmed」「search and log」「queries」時使用。
- - -

嚴格遵守以下規則：

1.【禁止改動】……
2.【紀錄 Log】……
3.【匯出結果檔】…… 路徑為：literature/results/<今天日期年月日>_S4.csv

Bonus: Automatic Script in Python

1. **Count hits**
   retmax=0 + count -> true total
2. **Fetch data**
   esummary + efetch
3. **Write the log**
   appends a row to search_log.md
4. **Export**
   CSV + abstracts.txt

In your research folder, run:

```
python demo.py
```

Returns:

Total fetch
Actual queries

- **MAX_RECORDS = 200** -> fetch cap; **MAX_RECORDS = 0** -> fetch ALL
- **API_KEY** can stay empty -> drops you from 10 requests/sec to 3, free at account.ncbi.nlm.nih.gov/settings/

Bonus: Automatic Script in Python

Find `pubmed_fetch.py`. In your folder command line, run:

```
python pubmed_fetch.py
```

```python
def count_hits(term):
    r = json.loads(_get("esearch.fcgi", {
        "db": "pubmed", "term": term,
        "retmode": "json", "retmax": 0
    }))["esearchresult"]
    return int(r["count"]), r["querytranslation"]
```

Still...

1. **Verify every citation**

   Does the PMID exist -> does the title match -> do the authors and year match?

2. **The data privacy line**

   Never upload un-deidentified records, clinical data, or unpublished datasets to any cloud AI tool

3. **AI organizes; you judge**

   ICMJE added "Part V: Use of Artificial Intelligence in Publishing" in January 2026. Check it before you submit



PubMed	pubmed.ncbi.nlm.nih.gov
MeSH Database	meshb.nlm.nih.gov
Search field tags	pubmed.ncbi.nlm.nih.gov/help/#using-search-field-tags
My NCBI (API key、alert)	account.ncbi.nlm.nih.gov/settings/
Gemini Notebook	notebook.google.com
Today's MCP server	npmjs.com/package/@ncukondo/mcp-server-pubmed



THE END!!

Questions?
Please reach out to
yienlyu@gmail.com

Yay thank you!